

Software Design Document (SDD)

[Project Name]

Version: [Version Number]

Date: [Date]

Author(s): [Author Name(s)]

1. Introduction

1.1 Purpose

Describe the objective of this document and its intended audience.

1.2 Scope

Provide a brief overview of the software, its main functionalities, and goals.

1.3 Intended Audience (e.g., developers, QA teams, project stakeholders, management, etc.).

1.4 Definitions and Acronyms

Define any specific terms or acronyms used in the document.

2. System Overview

Include Innovative Idea or Problem Statement here

2.1 System Description

Summary of the system's functionality and its primary components.

2.2 System Boundaries

- Are there items that are commonly expected in a Software type that you will not be including?
- What is your focus?
- Are there external systems/entities your software interacts with?

2.3 Key Functionalities

2.4 System Architecture

2.4.1 Design Rationale

2.4.1.1 Major decisions taken/choices made (including architectural styles and patterns used)

2.4.1.2 Why a decision was made.

- Prior assumptions before decision.
- Alternatives considered.
- Trade-offs and criteria analyzed to select one approach/choice/item and reject the other(s).

NOTES:

Reasons for decisions are likely to be obvious to the current design team, BUT can be less obvious to those who modify or maintain the system after deployment.

Recording the rationale enhances the software product's long-term maintainability. Continuing to capture the rationale for changes during maintenance also contributes to the software's viability.

A useful design outcome is insight into and explicit documentation of the major decisions taken, along with an explanation of the rationale for each decision.

- SWEBOK

2.4.2 High-level diagrams

- Illustrate the system's architecture and how components interact. (C4s Context, Container, Component diagrams)
- Containers and Components in your software might be influenced by your chosen architectural styles and patterns.

3. Design Considerations

3.1 Assumptions

List any assumptions made during the design process.

3.2 Constraints

Identify any limitations or constraints that impact the design (e.g., technical, time-based).

4. Detailed Design

4.1 Component Descriptions

- Detail each major component or module, including its purpose and functionality.
- Interface Design (APIs, inputs/outputs clearly stated)
- Error Handling Strategy

4.2 Data Structures

Describe key data structures, databases, or files used by the system.

4.3 User Interface

Provide sketches or descriptions of the main user interface elements.

5. Testing Plan

5.1 Testing Strategy

Outline the approach for testing the software, including types of tests (e.g., unit, integration).

Consult SWEBOK 4 for different types of testing.

5.2 Test Cases

Provide examples of specific scenarios to be tested.

6. Glossary

Define any additional terms or concepts relevant to the project.

References:

- <https://www.compucholar.com/schools/blog/project-documentation-templates>
- <https://www.atlassian.com/work-management/knowledge-sharing/documentation/software-design-document>